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A. Prompts

In this section, we illustrate the exact prompts used in our paper. Fig. 11 illustrated the prompt used for automatic GPT-4 evaluation, which we borrow directly from Lin et al. (2023). In Fig. 10, we illustrate the prompt used for our proposed method of simplifying responses in the IT dataset to reduce model halluicnations.

```
# Simplification Prompt
↳I want you to act as a Prompt Rewriter. I will provide you with an query-response pair and you will need to simplify the response.

##Query: {instruction}

##Response: {response}

Simplify the response to only contain enough information that is helpful and factual with respect to the given instruction. Remove redundant information that increases the depth of the answer, and you should rewrite the response with brevity. The simplified response should be concise and to the point. If the response cannot be simplified, for example, the response is already concise, or it is an instruction that asks to code or summarize, do not do anything with it. Do not change the formatting of the response.

##Simplified Response:
```

Figure 10. Prompt for response simplification in IT datasets.

```
# Evaluation Prompt
↳ Please act as an impartial judge and evaluate the quality of the responses provided. You will rate the quality of the output on multiple aspects, such as Helpfulness, Clarity, Factuality, Depth, and Engagement.

##Query: {instruction}

##Output: {prediction}

##Evaluate
### Aspects
- Helpfulness: Rate the response based on how well it addresses the users query and provides a relevant solution. A score of 5 indicates the answer fully aids the user, while a 1 suggests it offers little to no help.
- Clarity: Rate the response based on how well-structured it is, with ideas presented in a clear and coherent manner. A high score of 5 means the answer is clear and logically structured, while a 1 suggests a disjointed or confusing reply.
- Factuality: Evaluate the factual accuracy and truthfulness of the information provided. A perfect 5 indicates the information is entirely correct and accurate, while a 1 suggests it has significant factual errors.
- Depth: Determine the level of detail and thoroughness in the response. A score of 5 means the answer delves deeply into the topic, while a 1 indicates it barely scratches the surface.
- Engagement: Assess how engaging and natural the response sounds in a conversational context. A high score of 5 reflects a response that feels engaging and human-like in its tone, while a 1 indicates a robotic or boring reply. ,

### Format Given the query, please rate the quality of the output by scoring it from 1 to 5 , individually on **each aspect**.
- 1: strongly disagree
- 2: disagree
- 3: neutral
- 4: agree
- 5: strongly agree

Now, please output your scores and a short rationale below in a json format by filling in the placeholders in []: { 'helpfulness': { 'reason': '[your rationale]', 'score': '[score from 1 to 5]' }, 'clarity': { 'reason': '[your rationale]', 'score': '[score from 1 to 5]' }, 'factuality': { 'reason': '[your rationale]', 'score': '[score from 1 to 5]' }, 'depth': { 'reason': '[your rationale]', 'score': '[score from 1 to 5]' }, 'engagement': { 'reason': '[your rationale]', 'score': '[score from 1 to 5]' } }"
```

Figure 11. Prompt for multi-aspect evaluation of model responses on open-domain chat instructions borrowed from Lin et al. (2023).

B. Qualitative Examples

Table 1, 2, 3, 4 and 5 show examples of cases 1, 2, 3, 4, and 5 defined in Fig. 6. Precisely, we compare responses on just-eval-instruct $_{\mathbf{1k}}$, by LLaMa-2 $_{\mathbf{7B}}$ trained on LIMA $_{\mathbf{1K}}$ and LIMA-Simple $_{\mathbf{1k}}$. We provide details on how LIMA-Simple $_{\mathbf{1k}}$ was constructed in Section 4.

Table 6, 7 and 8 show examples of hallucinations by LLaMa-2 $_{\mathbf{7B}}$, fine-tuned on databricks-dolly $_{\mathbf{15k}}$ with SFT. We show hallucinations for responses on just-eval-instruct $_{\mathbf{1k}}$, and their exact origins, i.e., the IT instance from the IT dataset the model was fine-tuned on. We present a causal analysis of such hallucinations in Section 5, where we hypothesize that the model borrows these tokens incorrectly from IT instances with similar concepts. Additionally, we show two more instances from the IT dataset with similar concepts which the model could have borrowed the hallucinated tokens from. Through this, we show that among instances with similar concepts, the fine-tuned model may borrow tokens from any instance randomly. Finally, we also show the results of a counterfactual study by human annotators, who asked the same question to the model but in varied ways in an attempt to escape the causal borrowing of tokens. The counterfactual questions were designed with care to explicitly remove the causal concepts identified in the earlier stage. For example, in the 3rd example in Table 7, the annotators changed the concept of *creation* to include *credited* in the IT, leading to a correct answer. This further amplifies our hypothesis of the model borrowing tokens only from IT instances with similar *concepts*. For this study, we show both successful and failed attempts by human annotators.

Table 12 shows examples of hallucinations in non-information-seeking instructions, like procedural (e.g., summarization) and coding instructions. We highlight two specific points through examples in this table: (1) Causal links for hallucinations can even be seen in non-information-seeking instructions, where models borrow tokens inaccurately from the IT dataset to generate responses, and (2) For open-ended instructions, which do not have a definite answer, like the ones shown in the table, models tend to borrow tokens or facts learned from the IT dataset itself for responding. This can be noticed for accurate responses and not just for hallucinations.

Finally, we also compare perfectly accurate answers by models trained on different datasets in Table 9, where we show that models trained on different datasets can generate quite different stylistic answers for the same question and still be accurate. Through this, we can attribute IT as a *knowledge extractor*, and we leave further investigation as part of future work.

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1. Instruction: Who created the Superman cartoon character?

Res.: Superman was created by writer Jerry Siegel and artist Joe Shuster. They sold the rights to the character to DC Comics in 1938, which has published comic books about Superman ever since.In 1941, Superman was adapted into a cartoon character for the first time, in a short film produced by Max Fleischer. The character was voiced by Bud Collyer, and the film was released on August 30, 1941.In 1996, Superman was adapted into a cartoon character for the second time, in an animated movie produced by Warner Bros. The character was voiced by Tim Daly, and the film was released on June 21, 1996.

Simplified Res.: Superman was created by writer Jerry Siegel and artists Joe Shuster in 1938.

---

2. Instruction: What is the difference between HTML and JavaScript?

Res.: Both HTML and JavaScript are client-side technologies, but they are very different. HTML is a language for describing the structure of a web page. It is used by the browser to render the page. JavaScript is a language for executing code on the client side. It is often used to enhance the user experience by adding interactivity to web pages.In a few words: HTML is the what; JavaScript is the how.The most common use of JavaScript is to add event handlers to web pages. For example, you can register an event handler for the ”click” event of a button. When the button is clicked, your event handler function is called. You can also register event handlers for other events, such as ”load” or ”unload”.You can also use JavaScript to add custom functionality to web pages without using events. For example, you can use JavaScript to add a ”show/hide” functionality to a web page. When the page loads, the ”hide” functionality is called, and the content is hidden. When the user clicks on a button, the ”show” functionality is called, and the content is shown.

Simplified Res.: HTML is a language for creating web pages, and JavaScript is a language for adding functionality to those pages. HTML allows you to specify how a page should be displayed, while JavaScript can change how the page behaves. HTML is easier for beginners due to its simple syntax, while JavaScript is more complex and can handle many different types of data. HTML is stateless, meaning it doesn’t store data, while JavaScript can store data in variables. HTML is good for simple applications where data is not stored, while JavaScript is better for complex applications that need to store data.

Table 1. Comparison of LLaMa-2 $_{\mathbf{7B}}$_SFT_LIMA $_{\mathbf{1K}}$ and LLaMa-2 $_{\mathbf{7B}}$_SFT_LIMA-Simple $_{\mathbf{1k}}$. We show two examples of Case 1: Responses by both responses are accurate, but response by LLaMa-2 $_{\mathbf{7B}}$_SFT_LIMA $_{\mathbf{1K}}$ is more detailed and preferred over LLaMa-2 $_{\mathbf{7B}}$_SFT_LIMA-Simple $_{\mathbf{1k}}$.

1. Instruction: How do you become an author?

Res.: There is no one way to become an author. Some people are born with the talent and others develop it. Here are some steps you can take to become an author:1. Read a lot. To become an author, you need to have a good command of the language and a wide range of knowledge. The best way to get both is to read a lot. Read fiction and non-fiction books on a variety of subjects.2. Write a lot. The best way to become an author is to write a lot. Start with short stories and work your way up to novels. Write about things you know and things you want to learn about.3. Get feedback. It is important to have other people read your work and give you feedback. You can get feedback from friends, family, and even strangers online.4. Find your voice. It is important for authors to have a distinct voice. This means finding the style and tone that is unique to you.5. Consider your audience. When you are writing, keep your audience in mind. Will you be writing for children or adults? Do you want to write fiction or non-fiction?6. Set a schedule. It is important to stick to a schedule if you want to become an author. You should set aside time each day or each week to work on your writing.7. Be patient. It can take a long time to become an author. Don't get discouraged if your work isn't published right away.8. Keep learning. Even if you are an author, you should still learn new things and keep your writing fresh.9. Consider self-publishing. These days, it is very easy to self-publish a book. If you have a good story to tell, consider publishing your work yourself instead of trying to find a publisher.10. Get a mentor. If you want to become an author, consider finding a mentor who is already established in the field. A mentor can help you learn the ropes and give you advice along the way.

Simplified Res.: To become an author, you need to write a book. Here's how to do it:1. Choose Your Topic - Pick a topic you're interested in or know well. - Find out what's already been written on the topic to avoid duplication. - Consider writing about a personal experience or hobby. - Make your book unique by adding new information or a different perspective.2. Plan Your Book - Develop a clear outline of your book's content. - Identify the main "truths" you've found in your research. - Use a simple, clear writing style. - Avoid complex scientific or legal jargon. - Make your book fun and engaging.3. Write Your Book - Use notes with key points to make writing easier. - Focus on the main "truths" you've found. - Use simple, clear language. - Make your book engaging and fun.4. Edit Your Book - Ensure each chapter is clear and concise. - Remove unnecessary details and jargon. - Add more information as needed to support your main "truths."5. Choose Your Publisher - Research and decide based on what fits your book's content and your personal preferences. - Consider self-publishing if you want full control over your book. - Be aware of potential costs and time investments.6. Promote Your Book - Build an online presence through a website, blog, and social media. - Network with professionals in your industry. - Get book reviews from experts. - Consider paid advertising if you want to reach a larger audience.

2 Instruction: Did Facebook corporation change its name?

Res: Yes, Facebook rebranded itself as Meta in October 2021.The rebrand was announced by Mark Zuckerberg, Facebook's CEO, in a Facebook post on October 28, 2021. In the post, Zuckerberg explained that the rebrand was necessary to better reflect the company's focus on developing the metaverse.Prior to the rebrand, the company was known as Facebook Inc. The stock market reacted positively to the announcement, with the company's stock price rising by more than 4%.

Simplified Res.: Yes, Facebook changed its name to Meta in 2021. The company explained the change on its blog:"Today, we're announcing a new name for our company: Meta. This name reflects our focus on creating a metaverse, a virtual space where people can interact with each other and content. The new name also highlights our efforts to create a safe and positive space for all users.

Table 2. Comparison of LLaMa-2 $_{\mathbf{7B}}$_SFT_LIMA $_{\mathbf{1K}}$ and LLaMa-2 $_{\mathbf{7B}}$_SFT_LIMA-Simple $_{\mathbf{1k}}$. We show two examples of Case 2: Responses by both models are accurate but different information-wise or just a rephrased version of each other.

1. Instruction: How did US states get their names?
Res.: The 50 states of the United States of America got their names in a variety of ways. Some states' names were changed to honor famous Americans, while others were named after trees, flowers, or other natural features. Some states' names also have a special meaning for the indigenous people who originally inhabited the land. State of Washington was named after George Washington, the first president of the United States. State of Oregon was named after the river Orinoco, which was named after the city of Orl0̆0e9ans in France, the birthplace of French explorer Robert de La Salle, who explored the area in 1763. State of California was named after the Spanish missionary Franciscan friar Junoedpero Serra, who founded the first Spanish mission in the state in 1769. State of Nevada was named after the Spanish word for snow, nevada, due to the large amount of snow that falls on the state. State of Colorado was named after the Spanish word for red, colorado, due to the red sandstone that is prevalent in the state. State of New Mexico was named after the Spanish word for New Mexico, which was originally spelled "Nuevo M0̆0e9xico" in Spanish. State of Arizona was named after the Spanish word for the Phoenix, the mythical bird that is associated with the state. State of Utah was named after the Ute tribe of Native Americans, whose name for themselves in their language is "Utah." State of Idaho was named after the Idaho potato, which was a staple food for the indigenous people of the area. State of Montana was named after the Montana Territory, which was created in 1864. State of Washington D.C. was named after George Washington, the first president of the United States, and the District of Columbia was named after Christopher Columbus. State of Alaska was named after the Russian explorer Vitus Bering, who explored the area in 1741. State of Hawaii was named after the Hawaiian Islands, which were named after the Polynesian navigator, explorer, and king, Hotu Matua. State of Guam was named after the Spanish word for.
Simplified Res.: US states get their names from two sources:1. The first settlers named places after local Native American tribes.2. Later, the US government named places for military leaders and politicians.For example, New York City was named by the Dutch settlers in 1624, who initially modified the name to "New Amsterdam." The US government later changed the name to "New York" in 1845.Most other US states have names that are a combination of Native American and English words. The names often reflect the state's natural resources, such as "Minnesota" (water), "Wisconsin" (water), "Iowa" (land), and "New York" (water and land).
2 Instruction: What are some species of bears that are now extinct?
Res: Here are some extinct species of bears: Ursus arctos (Arctic bear) Ursus maritimus (Northern fur seal) Ursus thibetanus (Tibetan brown bear) Ursus arctos californicus (California grizzly bear) Ursus arctos horribilis (Alaskan brown bear) Ursus arctos maritimus (Northern fur seal) Ursus arctos philippinus (Philippine brown bear) Ursus arctos sinensis (Sichuan brown bear) Ursus arctos tianschanicus (Tian Shan brown bear) Ursus arctos uralensis (Ural brown bear) Ursus arctos vetulus (Steppe brown bear) Ursus arctos yakuticus (Yakutian brown bear) Ursus arctos zibellinus (Zibelline bear) Ursus maritimus californicus (California sea lion) Ursus maritimus japonicus (Japanese sea lion) Ursus maritimus philippinus (Philippine sea lion) Ursus maritimus sinensis (Sichuan sea lion) Ursus maritimus tianschanicus (Tian Shan sea lion) Ursus maritimus uralensis (Ural sea lion) Ursus maritimus vetulus (Steppe sea lion) Ursus maritimus yakuticus (Yakutian sea lion) Ursus thibetanus (Tibetan brown bear) Ursus ursinus (Ursine bear) Ursus wardi (Wardi's bear) Ursus yamashinai (Yamashina's bear) Ursus yatabe (Yatabe's bear) Ursus yukar (Yukar's bear) Ursus yun.
Simplified Res.: The following bears are now extinct: The European brown bear (Ursus arctos) The North American black bear (Ursus americanus) The Asian black bear (Ursus thibetanus) The Polar bear (Ursus maritimus) The Sun bear (Ursus helo) The Moon bear (Ursus lunaris) The Black-legged kittens (Ursus pardus) The Red-legged kittens (Ursus rubescens)

Table 3. Comparison of LLaMa-2 $_{\mathbf{7B}}$_SFT_LIMA $_{\mathbf{1K}}$ and LLaMa-2 $_{\mathbf{7B}}$_SFT_LIMA-Simple $_{\mathbf{1k}}$. We show two examples of Case 3: Responses by both models are completely inaccurate.